

Communicating Effectively with Visuals

Visuals are pervasive in today's media-rich society in both our work and leisure activities. We read and write documents and reports that include images; we hear and give presentations where the audience expects graphs and other visuals. In fact, each day millions of people are clicking on Web sites such as Flickr.com and YouTube.com where they view and post photos and videos. Additionally, other specialty sites, such as SlideShare.net, are becoming increasingly popular. Visuals are even being used as search expressions with smartphone apps, such as Google Goggles, for finding information. This growing use of images for conveying information confirms that being visually literate is extremely important in order to communicate effectively today.

One recent experimental study of business executives examined the impact of using visuals in group work. Its results clearly showed that groups who were using visuals in their work had higher productivity, higher quality of outcomes, and greater knowledge gains for the individuals in the group.¹ ■

LEARNING OBJECTIVES

- LO 3-1** Plan which parts of your document or presentation should be communicated or supported by visuals.
- LO 3-2** Explain the factors that are important in the effective presentation of visuals: size, layout, type, rules and borders, color and cross-hatching, clip art, background, numbering, titles, title placement, and footnotes and acknowledgments.
- LO 3-3** Construct textual visuals such as tables, pull quotes, flowcharts, and process charts.
- LO 3-4** Construct and use visuals such as bar charts, pie charts, line charts, scatter diagrams, and maps.
- LO 3-5** Avoid common errors and ethical problems when constructing and using visuals.
- LO 3-6** Place and interpret visuals effectively.

workplace scenario



Choosing the Right Visuals for Your Data and Your Audience

In your job as the sales manager for Green Living Industries, you are frequently called upon to gather sales data and market demographics. You share these data with your employees as well as your colleagues and corporate headquarters to support decisions that help achieve business goals.

In fact, you have just finished gathering data for a major report you will give orally

and in writing to Green Living's upper management. These data include information on the makeup of your target markets, sales performance by division and geographic area, industry trends, and customer perceptions of your products. Although your primary audience is Green Living's upper management, you also know that these data will be used to inform decisions at various district

offices and sales divisions. Essentially, you have a lot of information and several audiences who will use it. You need to think critically regarding how you will present this information so that your audiences quickly, clearly, and correctly understand what you are communicating.

Visual communication is the intentional use of illustrations (e.g., charts, pictures, maps, diagrams) and formatting (e.g., typography, color) to deliver a message. It may seem like you're simply putting numbers in a table, demonstrating a trend in a chart, or getting an audience's attention, but every visual you choose tells your story and leaves the audience with an impression of you, your data, and your ideas. In other words, visuals are never neutral. Rather, they are as essential to communicating meaning and presenting a professional image as the words written in a document or spoken in a presentation.

In addition, as a consumer of visual information, you will want to look critically at the visuals you encounter to ask what message is being conveyed and whether it is being conveyed clearly, accurately, and ethically.

LO 3-1 Plan which parts of your document or presentation should be communicated or supported by visuals.

PLANNING THE VISUALS

You should plan the visuals for a document soon after you organize your findings. Your planning for visuals should be guided largely by your business and communication goals. Visuals can clarify complex or difficult information, emphasize facts, add coherence, summarize data, set the tone of your discussion, inform or persuade an audience, and provide interest. Of course, well-constructed visuals also enhance your document's appearance.

In planning visuals, you should review the information that your document will contain, looking for any possibility of improving communication of the material through the use of visuals. Specifically, you should look for complex information that visual presentation can make clear, for information too detailed to be covered in words, and for information that deserves special emphasis.

Of course, you want to plan with your reader in mind and choose visuals appropriate to both the content and context where they are presented. You should construct visuals to help the intended reader understand the information more quickly, easily, and completely.

As you plan the visuals, remember that some visuals can stand alone, but others will supplement the writing or speaking—not take its place. Visuals in documents or oral presentations should support your words by communicating difficult concepts, emphasizing the important points, and presenting details.

LO 3-2 Explain the factors that are important in the effective presentation of visuals: size, layout, type, rules and borders, color and cross-hatching, clip art, background, numbering, titles, title placement, and footnotes and acknowledgments.

DETERMINING THE GENERAL MECHANICS OF CONSTRUCTION

In constructing visuals, you will make decisions regarding the various conventions for presenting them to business audiences. The most common conventions are summarized in the following paragraphs.

Size

One of the first steps you must take to construct a visual is determining its size. The size of a visual is determined by its contents and importance. If a visual is simple (with only two or three quantities), a quarter page might be enough. But if a visual must display complex or detailed information, a full page might be justified.

With extremely complex, involved information, you may need to use more than a full page. When you do, make certain that

this large page is inserted and folded so that the readers can open it easily. You may also consider including large or complex visuals in an appendix or attachment.

Orientation

You should determine the **orientation** of the visual by considering its size and contents. Sometimes a tall orientation (portrait) is the answer; sometimes the answer is a wide orientation (landscape). Simply consider the logical possibilities and select the one that is most easily read.

Type

The type used in visuals throughout a report should generally be consistent in terms of **style and font**. Style refers to the look of the type such as bold or italics; font refers to the look of the letters, such as those with or without feet (*serif* or *sans serif*, respectively). Be aware that even the design of the font you choose will convey a message, a message that should work with the text content and design.

Type size is another variable to watch. The size you choose should look appropriate in the context in which it is used. Your top priority when choosing type style, font, and size should always be readability.

Rules and Borders

You should use **rules (lines) and borders** when they improve the readability of the visual. Rules help distinguish one section or visual from another, while borders help separate visuals from the text. Keep in mind that rules can add clutter, so be sure to use them only when they will enhance the audience's understanding of your visual. And when using borders, be sure to place borders around visuals that occupy less than a full page. You also can place borders around full-page visuals, but such borders serve mostly a decorative function. Except in cases in which visuals simply will not fit into the normal page layout, you should not extend the borders of visuals beyond the normal page margins.

Color and Cross-Hatching

Color and cross-hatching, appropriately used, help readers see comparisons and distinctions (see Exhibit 3-1). In fact, research has found that color in visuals improves the comprehension, retention, and extraction of information. Also, color and cross-hatching can add to the attractiveness of the report. Because color is especially effective for this purpose, you should use it whenever practical and appropriate.

Clip Art

Today you can get good-looking clip art easily—so easily, in fact, that

communication matters

Communicating with Color

When constructing visuals, it is important to choose colors that are appropriate for the audience, context, and message. Colors have culture-specific meanings, so when choosing colors for a U.S. audience, you will want to be aware of the following associations.

Warm colors (e.g., red, orange, yellow) communicate passion, happiness, enthusiasm, energy, importance, or warning. Cool colors (e.g., green, blue, purple) provide a sense of calm, balance, harmony, and professionalism. Neutrals (e.g., black, brown, gray, beige, white) are conservative colors that convey power, sophistication, dependability, reliability, and formality.

However, shades of these colors are also important. Olive green, for instance, is associated with nature, while dark green indicates affluence. Likewise, bright blues can be seen as energizing, while navy is conservative and businesslike, and light purple is associated with romance while dark purple is associated with wealth and royalty. However, colors can also have negative connotations. For example, black is often associated with death, beige can be boring, and green can be associated with jealousy or envy.

As you can see, choosing the right color is important for setting an appropriate tone and conveying your message. Be sure you incorporate color decisions when you plan visuals for your document or presentation.

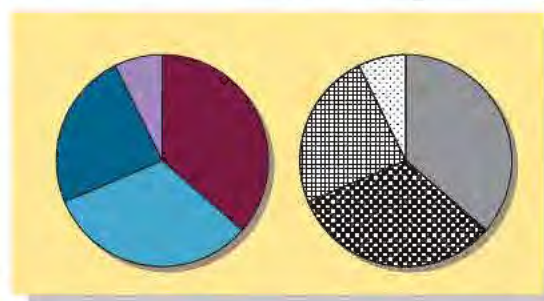
Source: Cameron Chapman, "Color Theory for Designers, Part 1: The Meaning of Color," *Smashing Magazine*, Smashing Media, 28 Jan. 2010, Web, 30 May 2013.

some writers often overuse it. Although clip art can add interest and bring the reader into a visual effectively, it also can overpower and distract the reader. The general rule is to keep in mind the purpose your clip art is serving: to help the reader understand the content. It should be appropriate in both its nature and size; and it should be appropriate in its representation of gender, race, and age. Also, if the clip art is copyrighted, you may need permission to use it.

Background

Background colors, photos, and art for your visuals should be chosen carefully. The color should provide

▼ EXHIBIT 3-1 Color versus Cross-Hatched Pie



communication matters

Applying Color to Visuals

As we discussed in the Communication Matters feature on page 43, colors help set your tone and communicate your message. While color selection itself is important, so, too, is applying color effectively to your visuals. Stephen Few provides the following rules for ensuring that color enhances your audience's understanding and your communication.

Rule #1: If you want different objects of the same color in a table or graph to look the same, make sure that the background—the color that surrounds them—is consistent.

Rule #2: If you want objects in a table or graph to be easily seen, use a background color that contrasts sufficiently with the object.

Rule #3: Use color only when needed to serve a particular communication goal. Color

should be used to enhance data; it should not be used for decoration. In many cases, less color may be better.

Rule #4: Use different colors only when they correspond to differences of meaning in the data. If audiences see similar units of data represented in different colors, they will assume the different colors have meaning, when really, the color was just for decoration.

Rule #5: Use soft, natural colors to display most information and bright and/or dark colors to highlight information that requires greater attention.

Rule #6: When using color to [a sequence of values, such as lowest to highest], stick with a single [color] (or a small set of closely related [colors]) and vary [shades] from pale

colors for low values to increasingly darker and brighter colors for high values.

Rule #7: Non-data components of tables and graphs should be displayed just visibly enough to perform their role, but no more so, for excessive salience could cause them to distract attention from the data. For example, use thin gray lines for borders.

Rule #8: To guarantee that most people who are colorblind can distinguish groups of data that are color coded, avoid using a combination of red and green in the same display.

Rule #9: Avoid using visual effects in graphs (e.g., shadows, 3-D effects).

Source: Practical Rules for Using Color in Charts, "Visual Business Intelligence Newsletter, Stephen Few, Perceptual Edge, February 2008. Reprinted with permission.

All visuals generally have titles; captions may be used if the reader will need them to interpret or better understand the visual.

high contrast with the data and not distract from the main message. As with any photos or art, backgrounds should create a positive, professional impression. Additionally, when visuals are used cross-culturally, you will want to be sure the message your background sends is the one you intended by testing or reviewing it with your audience.

Numbering

Pull quotes, clip art, and other decorative visuals do not need to be numbered. Neither does a lone table or figure in a document. Otherwise, you should number all the visuals. Many schemes of numbering are available to you, depending on the types of visuals you have.

If you have many visuals that fall into two or three categories, you may number each of the categories consecutively. For example, if your document is illustrated by six tables, five charts, and six maps, you may number these visuals Table I, Table II, ... Table VI; Chart 1, Chart 2, ... Chart 5; and Map 1, Map 2, ... Map 6.

However, if your visuals comprise a wide mixture of types, you may number them in two groups: tables and figures. Figures, a miscellaneous grouping, include all types other than tables. To illustrate, consider a report containing three tables, two maps, three charts, one diagram, and one photograph. You would number these visuals Table I, Table II, and Table III and Figure 1, Figure 2, ... Figure 7. Whatever your numbering scheme, remember that tables are numbered separately from other types of visuals.

Construction of Titles and Captions

The **title** is the name you give your visual; a **caption** is a brief description of the visual. All visuals generally have titles; captions may be used if the reader will need them to interpret or better understand the visual. Like the headings used in other parts of the report, the title or caption of the visual has the objective of concisely covering the contents. As a check of content coverage, you might use

communication matters

Infographics: Everything Old Is New Again

According to the folks at Visual.ly, the infographic has been around since humans used cave drawings. Even Florence Nightingale used an infographic (the Coxcomb chart) during the Crimean War to categorize diseases and causes of death. So while the infographic is nothing new, what is new is its popularity as a data visualization tool.

Though Visual.ly acknowledges that infographics serve much the same purpose as other visuals, infographics differ from other visuals in that they (1) “have a flow to them” that makes complex data more accessible to the reader, (2) present data so that the reader can engage with the story the data tell, and (3) are visually engaging illustrations of data stories.

If you search the Internet for tools to create infographics, you’ll find several at your disposal. You’ll also see you can create beautiful data visualizations on widely varying topics—everything from survey results to résumés. The visual to the right illustrates how data regarding the Professionalism in the Workplace study by the Center for

Professional Excellence at York College of Pennsylvania were captured clearly and concisely in an infographic. Should you incorporate infographics in your business writing or presentations? Like any type of visual, an infographic is only effective if it can tell a story more clearly than text alone. You’ll also want to keep in mind that some infographics can look informal or cartoonish. As in any business communication, you must analyze your audience, communication goals, and context to know if an infographic is appropriate.

Sources: “History of infographics,” Visual.ly, Visually, Inc., 2011, Web, 12 July 2012; “Professionalism in the Workplace by the Numbers,” *The Undercover Recruiter*, The Undercover Recruiter, n.d., Web, 31 May 2013; “What Is an infographic?,” Visual.ly, Visually, Inc., 2011, Web, 12 July 2012. Reprinted with permission of iProspect.



the journalist’s **five Ws**: *who*, *what*, *where*, *when*, and *why*. Sometimes you also might use *how*. But because conciseness is also important, it is not always necessary to include all the Ws in the title. For example, the title of a chart comparing the annual sales volume of the Texas and California territories of the Dell Company for the years 2010–2012 might be constructed as follows:

- Who: Dell Company
- What: Annual sales
- Where: Texas and California branches
- When: 2010–12
- Why: For comparison

The title might read, “Comparative Annual Sales of Texas and California Territories of the Dell Company, 2010–12.” For

even more conciseness, you could use a major title and subtitle. The major title might read, “A Texas and California Sales Comparison”; the subtitle might read, “Dell Company 2010–12.” Similarly, the caption might read “A Texas and California Sales Comparison: Dell Company 2010–12.”

Placement of Titles and Captions

Titles are placed above tables, but they may be placed either above or below other visuals. However, most software programs automatically place titles at the top. Captions, on the other hand, are generally placed below tables and other visuals. When typing titles and captions, use **title case** (the kind of capitalization used for book titles). For long captions, you may opt to use sentence case (capitalization of the first word and any proper nouns).

EXHIBIT 3-2 Good Arrangement of the Parts of a Typical Table

Table number and title	Table I—U.S. Internet User Penetration by Race/Ethnicity 2008–2014 (% of population in each group)							
Spanner heads		<i>Actual</i>			<i>Projected</i>			
Column heads		2008	2009	2010	2011	2012	2013	2014
Row heads	Non-Hispanic							
	White alone	72.0%	74.0%	76.1%	77.5%	79.0%	80.1%	81.2%
	Black alone	58.2%	60.5%	63.8%	66.9%	69.6%	71.7%	72.3%
	Asian alone	70.0%	71.2%	73.4%	75.5%	77.5%	79.5%	81.0%
	Other*	46.4%	50.0%	52.5%	55.0%	58.0%	62.0%	65.5%
	Hispanic**	53.5%	56.5%	59.5%	62.9%	65.0%	67.6%	70.0%
Footnote	*Includes native Americans, Alaska natives, Hawaiian and Pacific Islanders, and bi- and multiracial individuals. **Could be of any race.							
Source acknowledgment	SOURCE: eMarketer, March 2010. Source: Lisa E. Phillips "U.S. Internet User Penetration by Race/Ethnicity 2008–2014," <i>eMarketer Digital Intelligence</i> , eMarketer, Inc., 29 Sept. 2010, Web, 24 May 2013. Reprinted with permission of eMarketer, Inc.							

Footnotes and Acknowledgments

Parts of a visual sometimes require special explanation or elaboration. When this happens, you should use **footnotes**. These footnotes are concise explanations placed below the illustration by means of a superscript (raised) number or symbol (asterisk, dagger, double dagger, and so on) (see Exhibit 3-2). Footnotes are placed immediately below a visual with no caption. If a visual contains a caption, the footnote is placed below the caption.

Usually, a **source acknowledgment** is the bottom-most component of a visual. By *source acknowledgment* we mean a reference to the body or authority that deserves the credit for gathering the data used in the illustration. The entry consists of the word *Source* followed by a colon and the source information (in some cases, simply the source name will suffice). See Exhibit 3-2 for an illustration of source acknowledgment.

If you or your staff collected the data, you may either omit the source note or give the source as "Primary," in which case the note would read like this:

Source: Primary.

LO 3-3 Construct textual visuals such as tables, pull quotes, flowcharts, and process charts.

CONSTRUCTING TEXTUAL VISUALS

Visuals for communicating report information fall into two general categories: those that communicate primarily through textual content (words and numerals) and those that communicate

primarily through visual elements (charts and graphs). Included in the textual group are pull quotes and a variety of process charts (e.g., Gantt, flow, organization).

Tables

A **table** is an orderly arrangement of information in rows and columns. As we have noted, tables are textual visuals (not really pictures), but they communicate like visuals, and they have many of the characteristics of visuals.

Aside from the title, footnotes, and source designation previously discussed, a table contains heads, columns, and rows of data, as shown in Exhibit 3-2. Row heads are the titles of the rows of data, and column heads are the titles of the columns.

The construction of text tables is largely influenced by their purpose. Nevertheless, a few rules generally apply:

- If rows are long, the row heads may be repeated at the right.
- The em dash (—) or the abbreviation *n.a.* (or *N.A.* or *NA*), but not the zero, is used to indicate data not available.
- Since footnote numbers in a table full of numbers might be confusing, footnote references to numbers in the table should be keyed with asterisks (*), daggers (†), double daggers (‡), section marks (§), and so on. Small letters of the alphabet can be used when many references are made.
- Totals and subtotals should appear whenever they help readers interpret the table. The totals may be for each column and sometimes for each row. Row totals are usually placed at the right, but when they need emphasis, they may be placed at the left. Likewise, column totals are generally placed at the bottom of the column, but they may be placed at the top when the writer wants to emphasize them.

A ruled line (usually a double one) separates the totals from their components.

- The units used to represent the data must be clear. Unit descriptions (e.g., bushels, acres, pounds, dollars, or percentages) appear above the columns, as part of the headings or subheadings. If the data are in dollars, however, placing the dollar mark (\$) before the first entry in each column can be sufficient.

Tabular information need not always be presented in formal tables. In fact, short arrangements of data may be presented more effectively as parts of the text. Such arrangements are generally made as either leaderwork or text tabulations.

Leaderwork is the presentation of tabular material in the text without titles or rules. (*Leaders* are the repeated dots with intervening spaces.) Typically, a colon precedes the tabulation, as in this illustration:

The August sales of the representatives in the Western Region were as follows:

Charles B. Brown	\$33,517
Thelma Capp	39,703
Bill E. Knauth	38,198

EXHIBIT 3-3 Illustration of a Pull Quote

COVER STORY

It's a tragic tale — but not completely accurate, according to some tech-employment experts. The situation is more nuanced than what can be captured in a headline, and both workers and employers share responsibility for the gap, they say.

Most portentous, though, is that the gap, whatever its true nature, is rapidly becoming a yawning chasm — one that IT employees will have to cross sooner rather than later. Many hiring experts, IT managers and CIOs believe that the tech employment landscape will be radically different five years from now as more and more companies outsource IT operations to service providers, perhaps offshore, or move traditional IT jobs to other business units.

In the face of such rapid change, it's becoming clear that the one skill every member of the IT workforce needs is career management.

"Everybody is a free agent, navigating the corporate chaos," says Todd Weinman, president of The Weinman Group, an executive search firm headquartered in Oakland, Calif., that specializes in audit and corporate governance. In the IT job market, he says, "the people who are faring a little bit better are constantly cultivating their careers on a variety of fronts."

Tech employees log long hours, meaning they get a lot of hands-on experience, but they're not getting the training and other types of enrichment they need to develop their careers. "In addition to your 50-plus hours a week, you need in-depth coursework to refresh your skills, plus studying to sit for certifications," says Weinman. At many companies, employees used to be able to take time for those types of pursuits during the workday, but not anymore.

"Those who want to stay relevant have to work very hard" — at work and during off-hours, says Weinman, who is a member of the ISACA Leadership Development Committee. ISACA is an IT professional association that, among other things, provides security certifications

services, concurs. "In today's marketplace, if you have good references and a strong technical skill set and can communicate how you'll provide ROI, four jobs will be waiting for you," he says.

What amazes, and to some degree frustrates, Cullen are those instances when clients choose not to hire a job applicant because they can't check every box on their wish lists. "We're seeing this huge pent-up demand, and the pool of labor isn't growing. And yet, what's perplexing is just how specific hiring managers still are," he says. "They want this skill, that particular work on the network side, certifications, this many years of experience. Companies are not willing to take a risk. Nobody's jumping out the window to hire the average employee."

Weinman blames the Great Recession for starting IT down the path that led to the skills gap, while cautioning that an improved economy won't much ease the crunch for many workers.

"Companies are getting leaner and leaner. Starting in 2008, they downsized and streamlined, and they haven't replaced those positions," he observes. "If you're the hiring director of one of these very lean teams, you want only A+ workers. In the past, someone could get away with being a solid middle-of-the-road employee. Not anymore."

Charles Williams sees the situation from both sides. As manager of data systems at Georgia System Operations, an electric utility in Tucker, Ga., he wants and expects the people who report to him (currently there are seven) to keep their skills up to date. At the same time, he acknowledges that he is challenged to keep his own knowledge fresh when day-to-day duties take priority over opportunities to investigate up-and-coming technologies.

"In a way, it's natural for a manager to develop a technical skills gap. We're not able to sit down and play with things the way our employees might," he says. And that worries him: "I feel like I need

Text tabulations are simple tables, usually with column heads and sometimes with rules and borders. But they are not numbered, and they have no titles. They are made to read with the text, as in this example:

In August the sales of the representatives in the Western Region increased sharply from those for the preceding month, as these figures show:

Representative	July Sales	August Sales	Increase
Charles B. Brown	\$32,819	\$33,517	\$ 698
Thelma Capp	37,225	39,703	2,478
Bill E. Knauth	36,838	38,198	1,360

Pull Quotes

The **pull quote** is a textual visual that is often overlooked yet extremely useful for emphasizing key points. It is also useful when the text or content of the report does not lend itself naturally or easily to other visuals. By selecting a key sentence, copying it to a text box, enlarging it, and perhaps even enhancing it with a new font, style, or color, a writer can break up the visual boredom of a full page or screen of text.

Software lets users easily wrap text around shapes as well as along curves and irregular lines. Exhibit 3-3 shows an example that is simple yet effective in both drawing the reader's attention to a key point and adding visual interest to a page.

Bulleted Lists

Bulleted lists are listings of points arranged with bullets (•) to set them off. These lists can have a title that covers all the points, or they can appear without titles, as they appear in various places in this book. When you use this arrangement, make the points grammatically parallel. If the points have subparts, use **sub-bullets** for them. Make the sub-bullets different by color, size, shape, or weight. Darts, check marks, squares, or triangles can be used for the secondary bullets.

Flowcharts and Process Charts

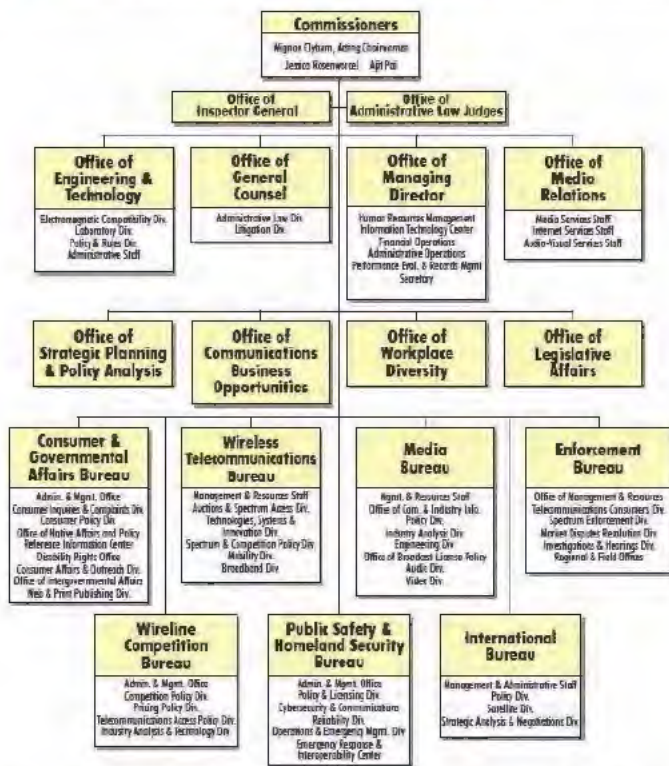
Business professionals use a variety of specialized charts in their work. Often these charts are a part of the

Companies are not willing to take a risk. Nobody's jumping out the window to hire the average employee.

JACK CULLEN, PRESIDENT, MOD 5

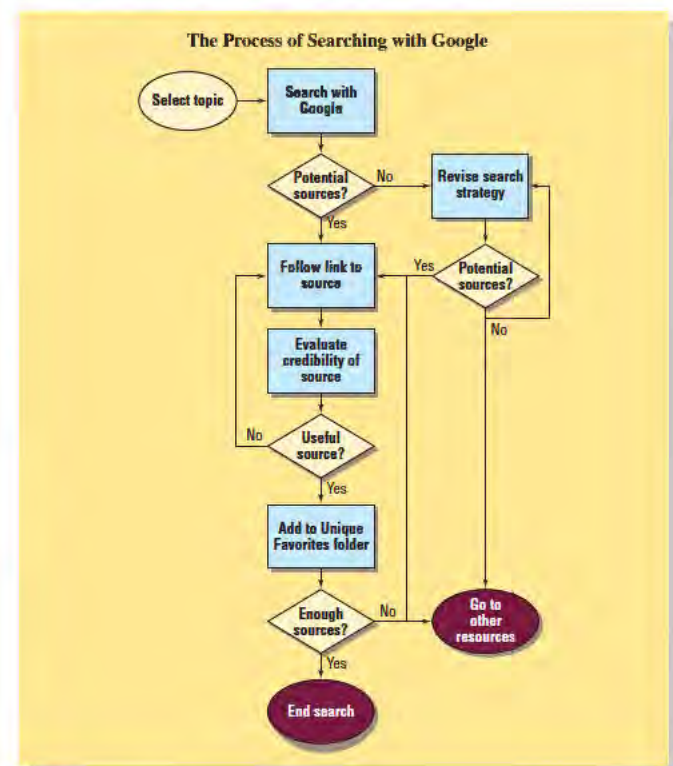
Source: Julia King and Tracy Mayor, "IT Skills: Jumping the Chasm," *Computerworld*, June 4, 2012, p. 19, print. Used with permission of Computerworld Online. Copyright © 2013. All rights reserved.

EXHIBIT 3-4 Illustration of an Organization Chart



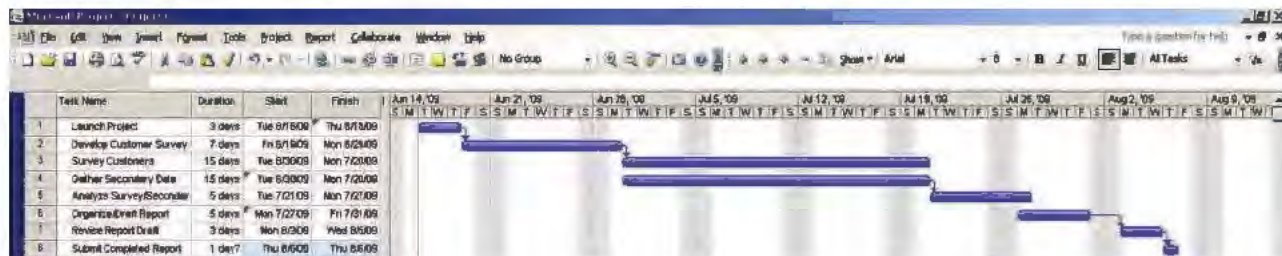
Source: "FCC Organizational Chart," FCC, Federal Communications Commission, May 2013, Web, 24 May 2013.

EXHIBIT 3-5 Illustration of a Flowchart



Source: Primary.

EXHIBIT 3-6 Illustration of a Gantt Chart



Source: Primary.

information presented in reports. Perhaps the most common of these is the **organization chart** (see Exhibit 3-4). This type of chart shows the hierarchy of levels and positions in an organization. A **flowchart** (see Exhibit 3-5), as the word implies, shows the sequence of activities in a process. Flowcharts use specific designs and symbols to show process paths. A variation of the organization and flowchart is the **decision tree**. This chart helps one follow a path to an appropriate decision. **Gantt charts** are visual presentations that show planning and scheduling activities (see Exhibit 3-6). You can easily construct these charts in a variety of applications.

LO 3-4 Construct and use visuals such as bar charts, pie charts, line charts, scatter diagrams, and maps.

CONSTRUCTING CHARTS, GRAPHS, AND OTHER VISUALS

Visuals built with raw data include bar, pie, and line charts and all their variations and combinations. Illustrations include maps, diagrams, drawings, photos, and cartoons.

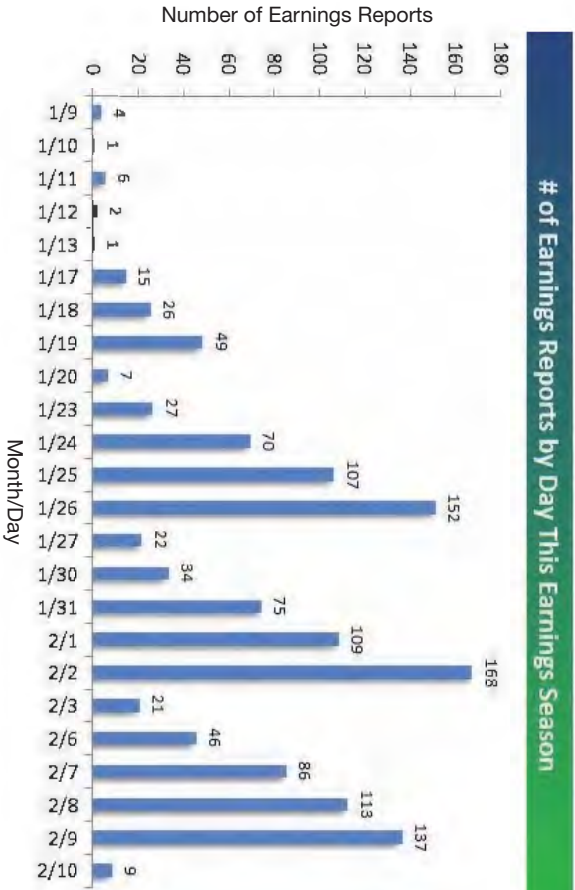
Bar and Column Charts

Simple bar and **column charts** compare differences in quantities using differences in the lengths of the bars to represent those quantities. You should use them primarily to show comparisons of qualities at a moment in time.

As shown in Exhibit 3-7, the main parts of the bar chart are the bars and the grid (the field on which the bars are placed). The bars, which may be arranged horizontally or vertically (then called a column chart), should be of equal width. You should identify each bar or column, usually with a caption at the left or bottom. The grid (field) on which the bars are placed is usually needed to show the magnitudes of the bars, and the units (e.g., dollars, pounds, miles) are identified by the scale caption below. It is often a good idea to include the numerical value represented by each bar for easy and precise comprehension, as shown in Exhibits 3-7 and 3-8.

When you need to compare quantities of two or three different values in one chart, you can use a **clustered** (or **multiple**) **bar chart**. Cross-hatching, colors, or other formatting on the bars distinguish the different kinds of information (see Exhibit 3-8). Somewhere within the chart, a **legend** explains what the different bars mean. Because clustered bar charts can become cluttered, you should usually limit comparisons to three to five kinds of information in one of them.

EXHIBIT 3-7 Illustration of a Bar Chart



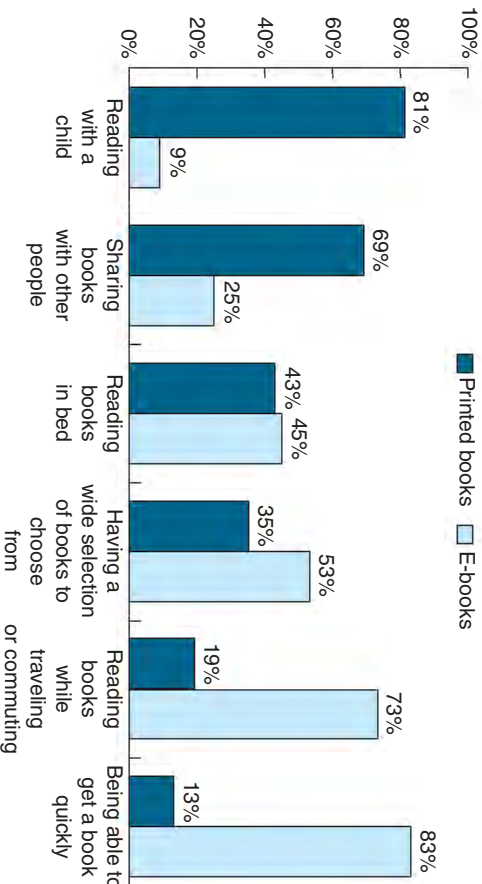
Source: "Earnings Reports by Day This Season," *Think B.I.G.* Bespoke Investment Group, 9 Jan. 2012, Web, 24 May 2013. <http://www.bespokeinvest.com/thinkbig/2012/1/9/earnings-reports-by-day-this-season.html>

When you need to show plus and minus differences, you can use **bilateral column charts**. The columns of these charts begin at a central point of reference and may go either up or down, as illustrated in Exhibit 3-9. Bar titles appear either within, above, or below the bars, depending on which placement works best. Bilateral column charts are especially good for showing percentage changes, but you may use them for any series that includes plus and minus quantities.

EXHIBIT 3-8 Illustration of a Clustered Bar Chart

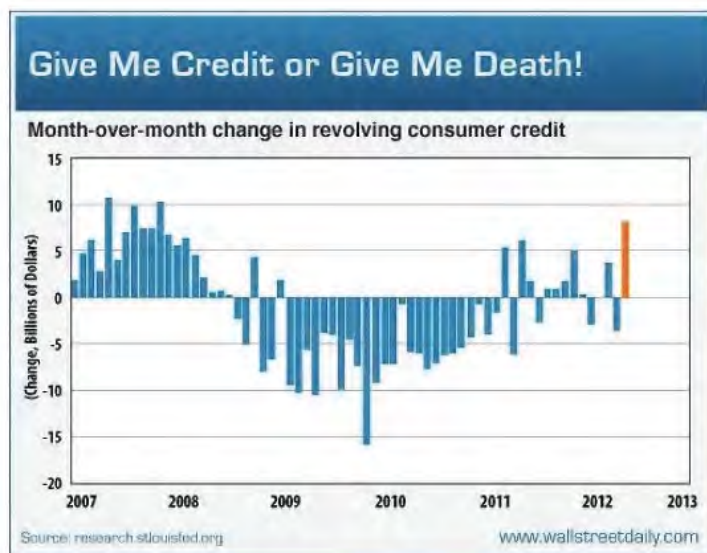
Which is better for these purposes, a printed book or an e-book?

% of those who have read both e-books and printed books in the last 12 months who say that this format is better for these purposes



Source: Lee Rainie, et al., "The Rise of e-Reading," *Pew Internet & American Life Project*, Pew Research Center, 4 Apr. 2012, Web, 24 May 2013.

EXHIBIT 3-9 Illustration of a Bilateral Column Chart



Source: Louis Basenese, "Friday Charts: Beer, Earnings, Recessions and Peak Oil Nonsense," *Wall St. Daily*, Wall Street Daily, LLC, 13 July 2012, Web, 24 May 2013. Reprinted with permission of Wall Street Daily.

If you need to compare subdivisions of columns, you can use a **stacked (subdivided) column chart**. Such a chart divides each column into its parts. It distinguishes these parts with color, cross-hatching, or other formatting; and it explains these differences in a legend. Subdivided columns may be difficult for your reader to interpret since both the beginning and ending points need to be found. Then the

reader has to subtract to find the size of the column component. Clustered bar charts or pie charts avoid this possibility for error.

Another feature that can lead to reader error in interpreting bar and column chart data is the use of three dimensions when only two variables are being compared. Therefore, unless more than two variables are used, choosing the two-dimensional presentation over the three-dimensional form is usually better. (Exhibit 3-20 illustrates the appropriate use of a three-dimensional visual to compare three dimensions.)

A special form of stacked (subdivided) column chart is used to compare the subdivisions of percentages. In this form, all the bars are equal in length because each represents 100 percent. Only the subdivisions within the bars vary. The objective of this form is to compare differences in how wholes are divided. The component parts may be labeled, as in Exhibit 3-11, or explained in a legend.

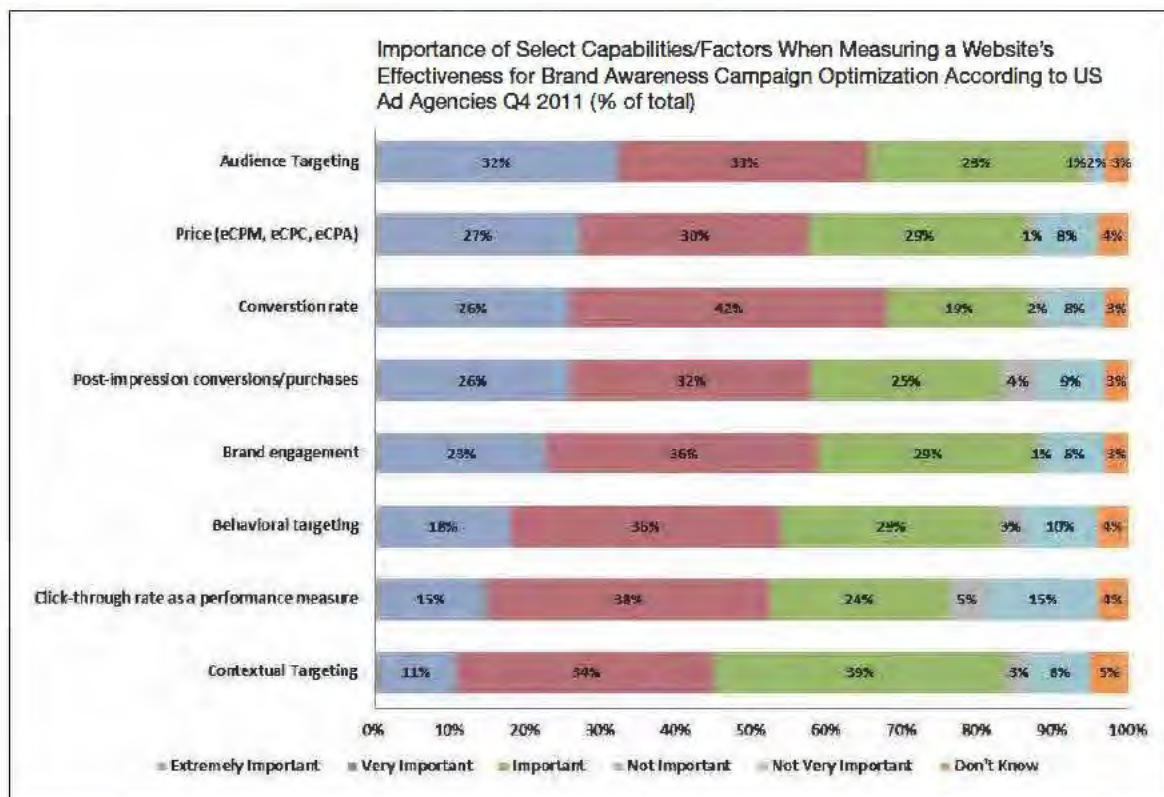
Pictographs

A **pictograph** is a bar or column chart that uses bars made of pictures. The pictures are typically drawings of the items being compared. For example, the number of senators in Exhibit 3-12 is represented by the image of a single person for each senator instead of by ordinary bars.

In constructing a pictograph, you should follow the procedures you used in constructing bar and column charts. In addition, you must make all the picture units equal in size. The human eye cannot accurately compare geometric designs that vary in

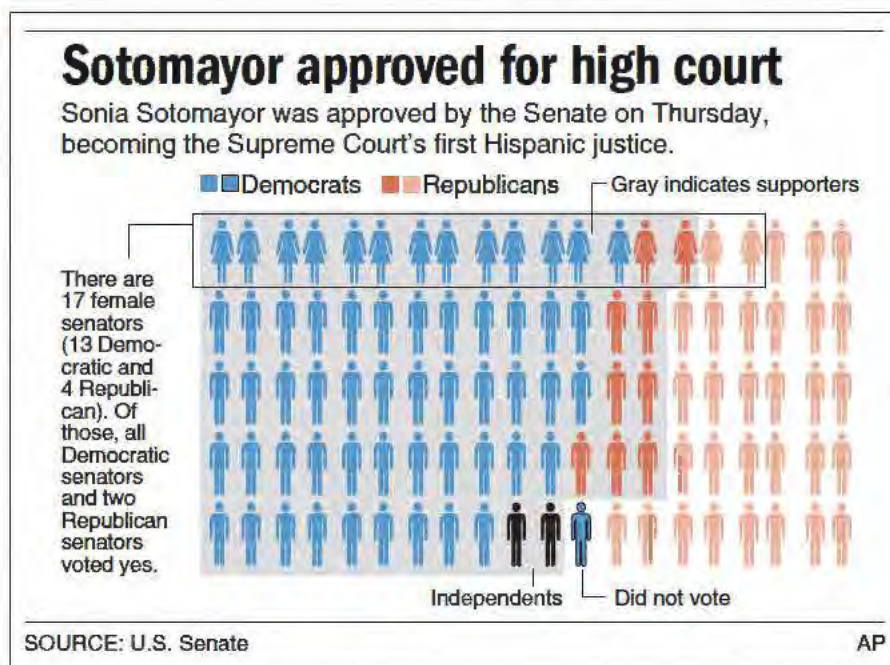


▼ **EXHIBIT 3-11** Illustration of a 100 Percent Stacked Column Chart



Source: Data from Maxifier, "Optimization Research," eMarketer, eMarketer, Inc., 14 Feb. 2012, Web, 24 May 2013. Reprinted with permission of eMarketer, Inc.

▼ **EXHIBIT 3-12** Illustration of a Pictograph



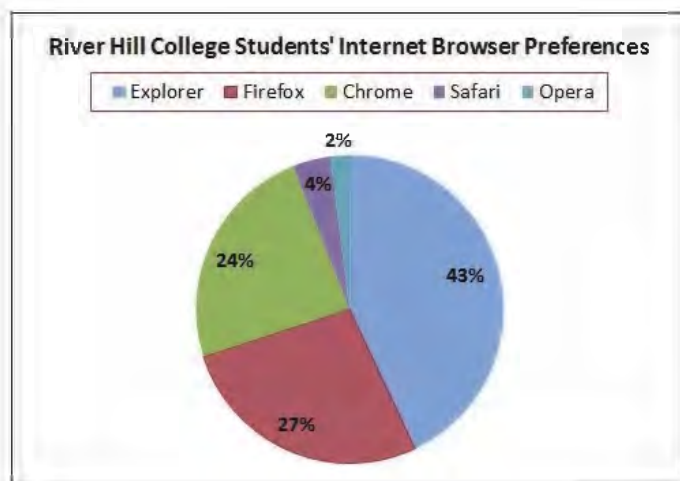
Source: D. Morris, "Sotomayor Approved for High Court," *The Washington Post*, The Washington Post Company, 7 Aug. 2009, Web, 24 May 2013. Reprinted with permission of AP Images.

more than one dimension, so show differences by varying the number, not the size, of the picture units. Also, be sure you select pictures or symbols that fit the information to be illustrated. In comparing the cruise lines of the world, for example, you might use ships. In comparing computers used in the world's major countries, you might use computers. The meaning of the drawings you use must be immediately clear to the readers.

Pie Charts

The most frequently used chart in comparing the subdivisions of wholes is the **pie chart** (see Exhibit 3-13). As the name implies, pie charts show the whole of the information being studied as a pie (circle) and the parts of this whole as slices of the pie. The slices may be distinguished by labeling and color or cross-hatching. A single slice can be emphasized by pulling it out from the pie or enlarging it. Because it is hard to judge the values of the slices with the naked eye, it is good to include the percentage values within or near

EXHIBIT 3-13 Illustration of a Pie Chart



Source: Primary.

each slice. Also, placing a label near each slice lets the reader more quickly understand the items being compared than using a legend to identify components. A good rule to follow for ordering the pieces of the pie is to begin by slicing the largest piece at the 12 o'clock position and then ordering the pieces largest to smallest; however, you should order the pieces in the way that will make the most visual sense to your audience.

Line Charts

Line charts are useful for showing changes of information over time. For example, changes in prices, sales totals, employment,

or production over a period of years can be shown well in a line chart.

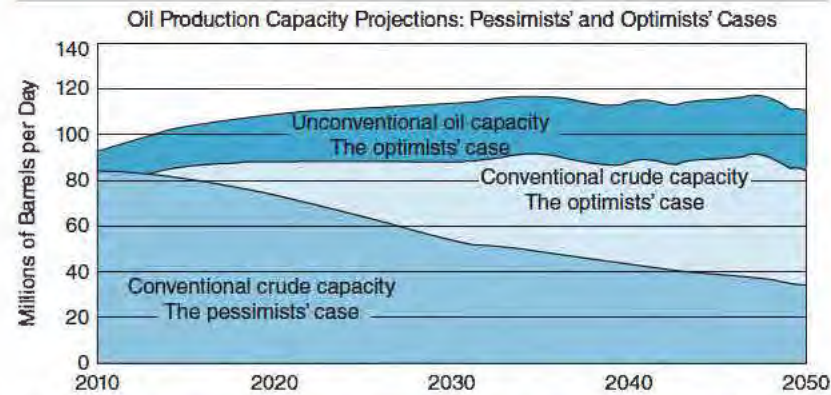
In constructing a line chart, draw the information to be illustrated as a continuous line on a grid that is scaled to show time changes from left to right (the X-axis) and quantity changes from bottom to top (the Y-axis). You should clearly mark the scale values and the time periods. They should be in equal increments.

You also may compare two or more series on the same line chart. In such a comparison, you should clearly distinguish the lines by color or form (e.g., dots, dashes, dots and dashes). You should also label them on the chart or with a legend somewhere in the chart. But the number of series that you can effectively compare on one line chart is limited. As a practical guide, the maximum number is five.

It is also possible to show parts of a series using an **area** chart. Such a chart, though, can show only one series. You should construct this type of chart, as shown in Exhibit 3-15, with a top line representing the total of the series. Then, starting from the base, you should cumulate the parts, beginning with the largest and ending with the smallest or beginning with the smallest and ending with the largest. You may use cross-hatching or coloring to distinguish the parts.

Line charts that show a range of data for particular times are called *variance* or *hi-lo* charts. Some variance charts show high and low points as well as the mean, median, or mode. When used to chart daily stock prices, they typically include closing price in addition to the high and low. When you use

EXHIBIT 3-15 Illustration of an Area Chart

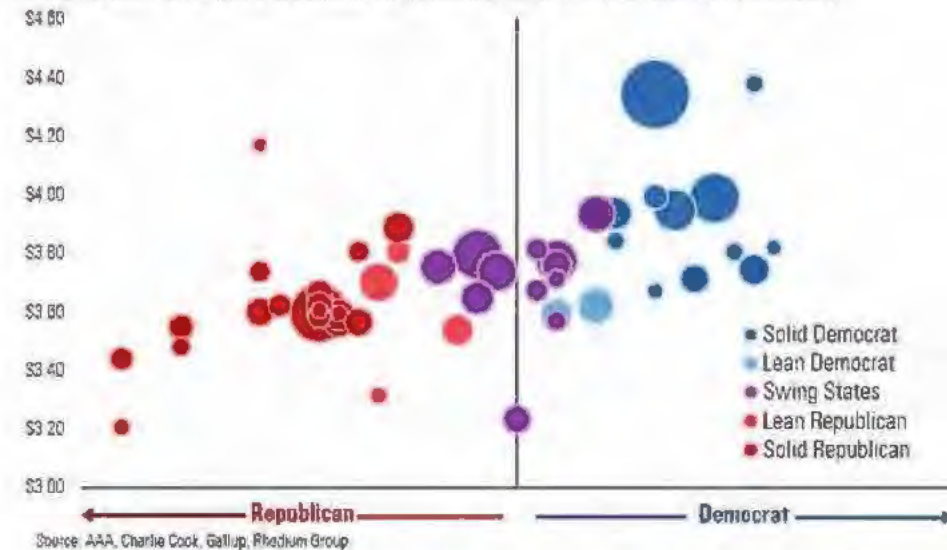


Source: Adapted with special permission from "The Argument for and Against Oil Abundance," *Bloomberg BusinessWeek*, 18 January 2010, p. 48.

EXHIBIT 3-16 Illustration of a Scatter Diagram

The Bluer the State, the Higher the Price

Average gasoline price (y-axis), partisan voting index (x-axis), and electoral votes (circle size)



Source: Trevor Houser, "Gasoline Prices and Electoral Politics in the Age of Unconventional Oil," *Notes*, Rhodium Group, LLC, 7 Mar. 2012, Web, 24 May 2013. <http://rhg.com/notes/gasoline-prices-and-electoral-politics-in-the-age-of-unconventional-oil>. Reprinted with permission.

points other than high and low, be sure to make it clear what these points are.

Scatter Diagrams

Scatter diagrams are often considered another variation of the line chart. Although they do use X- and Y-axes to plot paired values, the points stand alone without a line drawn through them. For example, a writer might use a scatter diagram in a report on digital cameras to plot values for price and resolution of several cameras. While clustering the points allows users to validate hunches about cause and effect, they can only be interpreted for correlation—the direction and strength

relationships. The points can reveal positive, negative, or no relationships. Additionally, by examining the tightness of the points, the user can see the strength of the relationship. The closer the points are to a straight line, the stronger the relationship. In Exhibit 3-16, the paired values are gas prices and political party affiliation.

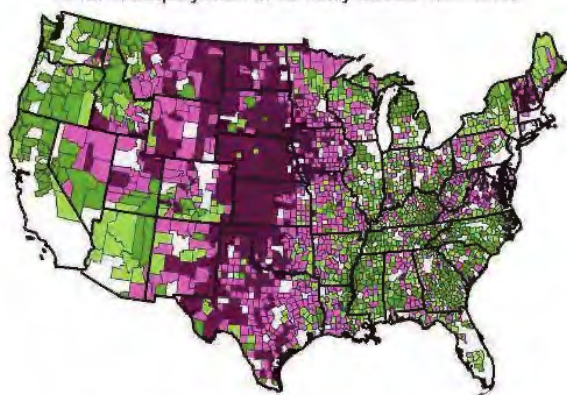
Maps

You also may use maps to communicate quantitative as well as physical (or geographic) information. **Statistical maps** are useful primarily when quantitative information is to be compared by geographic areas. On such maps, the geographic areas

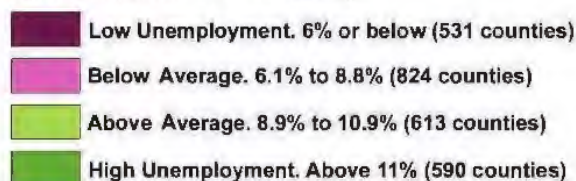
EXHIBIT 3-17 Illustration of a Map (Quantitative)

Employed Down the Middle

Rural unemployment in January divides the nation



Unemployment in rural and exurban counties in January 2012
National rate was 8.8%; rate in rural counties was 9.1%
(Blank areas above are metro regions.)



Source: "International, U.S. and Local Economic Indicators for May 2012," Scott Financial Group, 29 Apr. 2012, Web, 24 May 2013. <http://www.samuelscottfg.com/domestic-and-international-economic-indicators-for-may-2012>

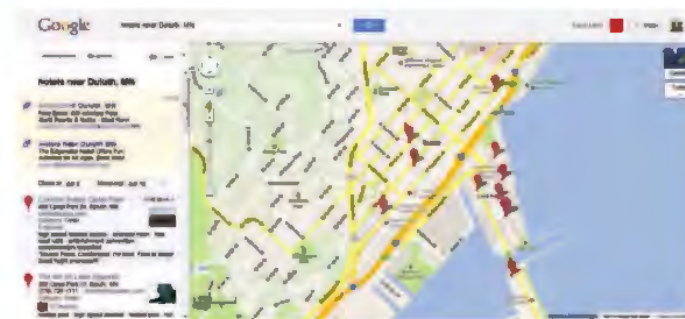
are clearly outlined, and formatting techniques are used to show the differences between areas (see Exhibit 3-17). Statistical maps are particularly useful in illustrating and analyzing complex data. **Physical or geographic** maps (see Exhibit 3-18) can show distributions as well as specific locations. Of the numerous formatting techniques available to you, these are the most common:

- Showing different areas with color, shading, or cross-hatching (see Exhibit 3-17). Maps using this technique must have a legend to explain the quantitative meanings of the various colors or cross-hatchings.
- Placing visuals, symbols, or clip art within each geographic area to depict the quantity for that area or geographic location.
- Placing the quantities in numerical form within each geographic area.

Combination Charts

Combination charts often serve readers extremely well by allowing them to see relationships of different kinds of data. The

EXHIBIT 3-18 Illustration of a Map (Physical)



Source: "Google Maps," Google.com. Google, 28 May 2013, Web, 28 May 2013. Google and the Google logo are registered trademarks of Google Inc., used with permission.

example in Exhibit 3-19 shows the reader the price of stock over time (the trend), the volume of sales over time (comparisons), and the MACD (an indicator of trends in stock performance). It allows the reader to detect whether the change in volume affects the price of the stock. This kind of information would be difficult to get from raw data alone.

Three-Dimensional Visuals

Earlier we said that **three-dimensional graphs** are generally undesirable. However, we have mostly been referring to the three-dimensional presentation of visuals with two variables. But when you actually have three or more variables, presenting them in three dimensions is an option if doing so will help your readers see the data from multiple perspectives and gain additional information. In fact, Francis Crick, who won a Nobel prize for discovering the structure of DNA, once revealed that he and his collaborators understood the configuration of DNA only when they took a sheet of paper, cut it, and twisted it. Today we have sophisticated statistics, visuals, and data-mining tools to help us see our data from multiple perspectives.

EXHIBIT 3-19 Illustration of a Combination Chart



Source: Reprinted with permission from Yahoo! Inc. © 2013 Yahoo! Inc. YAHOO! and the YAHOO! Logo are trademarks of Yahoo! Inc.



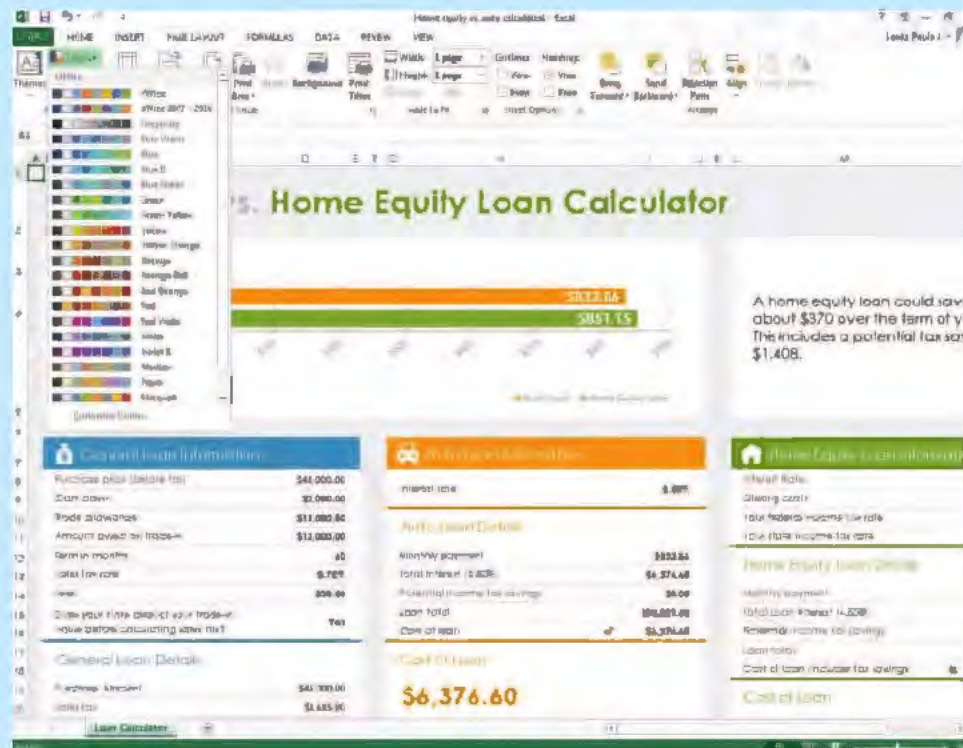
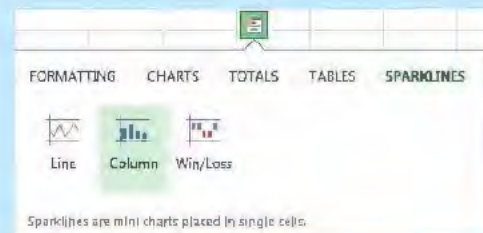
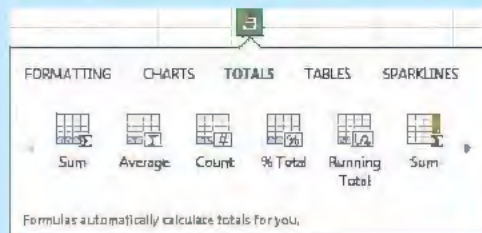
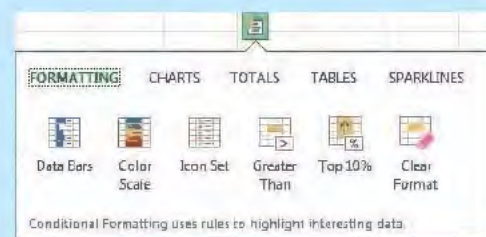
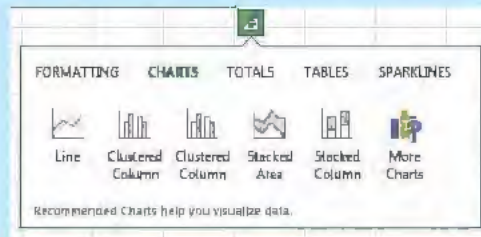
from the tech desk

Making the Most of Excel

Microsoft recently released Excel 2013. Though earlier versions of Excel were powerful, Excel 2013 presents even more exciting options for data visualization—and the best part is that you don't have to know a lot about

Excel to use them. Among other things, Excel 2013 suggests visuals appropriate for your data, provides numerous templates to ensure your data look sharp, and offers a Quick Analysis feature, which, as its name indicates, lets

you quickly calculate data and create visuals. A sample of a template and color palette options, and the Quick Analysis feature, appear here.



These three-dimensional tools are making their way from science labs into business settings. Thanks to the increased use of data-gathering tools, businesses large and small are collecting and attempting to analyze extremely large amounts of detailed data. They are analyzing not only their own data but also data on their competitors. And advances in hardware, software, and Web-based applications are making it easier to visually represent both quantitative and qualitative data.

Although 3D visuals help writers display the results of their data analysis, they change how readers look at information and may take some time to get used to. These tools enable users both to see data from new perspectives and to interact with them. They allow users to free themselves from two dimensions and give them ways to stretch their insights and see new possibilities.

Exhibit 3-20 shows a three-dimensional visual plot of factors identified as the major ones consumers use when deciding which slate computer to purchase. Five products are plotted on three variables: cost, battery life, and slate weight. This visual could help a company identify its major competitors and help consumers identify those products that are best suited to their needs. The more products or data that are plotted, the more valuable the graph is at helping the reader extract meaning. If these data had been displayed on a two-dimensional graph, the lines would have overlapped too much to be distinguishable, thus requiring you to accompany this visual with a table,

enable the reader to rotate the visual, or both in order to make the data clear.

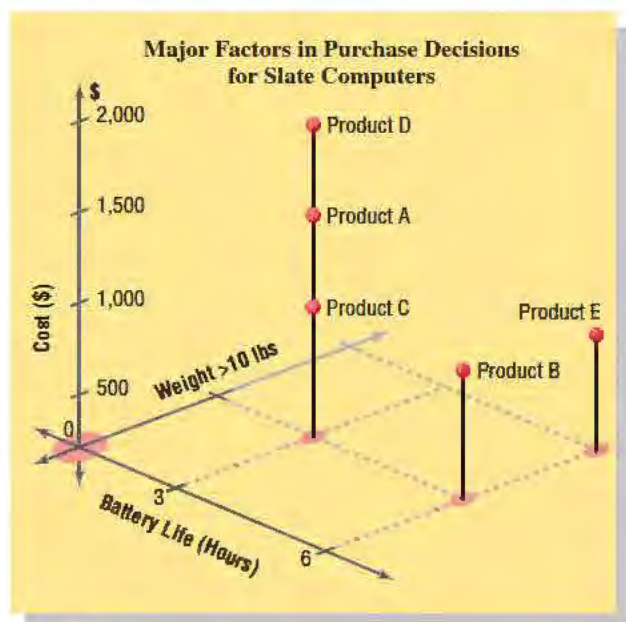
In deciding whether to use a three-dimensional representation or a two-dimensional one, you need to consider your audience, the context, and the goal of your communication. Overall, multidimensional presentation on paper is difficult; multiple representations can be made from separate two-dimensional views, but not always effectively. Moreover, if the 3D visual is being presented online or digitally where the reader can rotate it to see perspectives, it is likely to be much more effective.

Photographs

Cameras are everywhere today, enabling anyone to capture the images he or she needs. And royalty-free photos intended for commercial use are readily available on the Internet, too. Photos can serve useful communication purposes. They can be used to document events as well as show products, processes, or services. Exhibit 3-21 illustrates how a photo creates a message. What does the context created by the split image (half fall/half summer) suggest to you?

Today photos, like data-generated visuals, can easily be manipulated. A writer's job is to use them ethically, including getting permission when needed.

▼ **EXHIBIT 3-20** Illustration of a Three-Dimensional Visual



Source: Primary.

▼ **EXHIBIT 3-21** Illustration of a Photo





Other Visuals

The types of visuals discussed thus far are the ones most commonly used in business. Other types also may be helpful. **Diagrams** and drawings may help simplify a complicated explanation or description. **Icons** are another useful type of visual. You can create new icons, or you can select one from an existing body of icons with easily recognized meanings, such as ☹. Even carefully selected **cartoons** can be used effectively. **Video clips** and **animation** are now used in many electronic documents. For all practical purposes, any visual is acceptable as long as it helps communicate the intended story.

LO 3-5 Avoid common errors and ethical problems when constructing and using visuals.

Visual Integrity

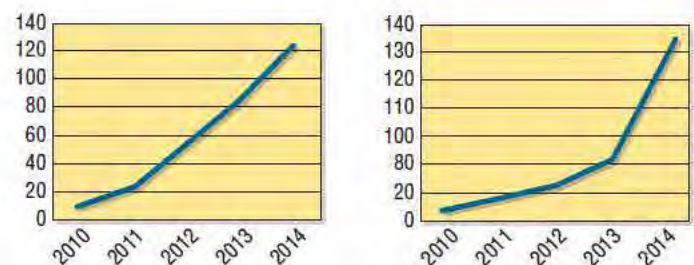
In writing a business document, you are ethically bound to present data and visuals in ways that enable readers to interpret them easily and accurately. By being aware of some of the common errors made in presenting visuals, you learn how to avoid them and how to spot them in other documents. Keep in mind that any errors—deliberate or not—compromise your credibility, casting doubt on the document as well as on other work you

have completed. Therefore, writers need to ensure that visuals accurately and honestly represent the data they contain.

avoiding errors in graphing data Common graphing errors are errors of scale and errors of format. Another category of error is inaccurate or misleading presentation of context.

Errors of scale occur whenever the dimensions from left to right (X-axis) or bottom to top (Y-axis) are unequal. Three sources of scale errors include no uniform scale size, scale distortion, and violating the zero beginning.

No uniform scale size occurs when intervals in data points on the X- or Y-axis are not consistent. Note in the graphs below how the unequal intervals on the Y-axis in the graph on the right create a different presentation of the data.



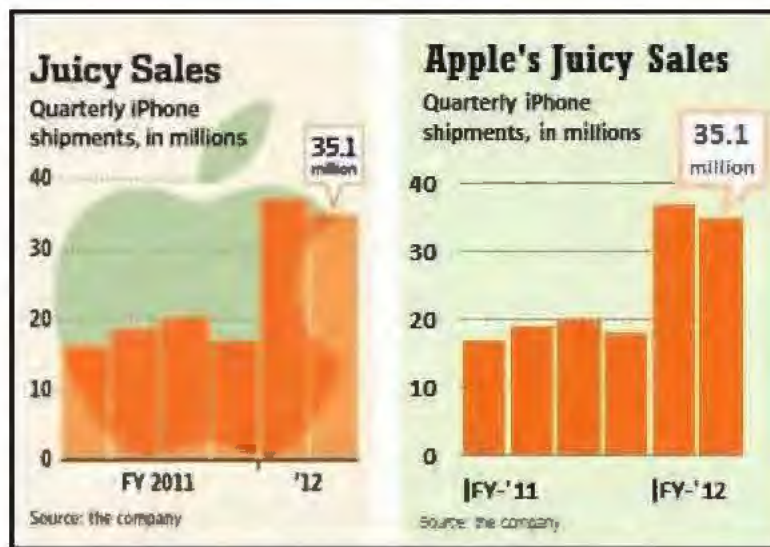
communication matters

Avoiding Chartjunk

In 1983 Edward Tufte, a pioneer and leading expert in data visualization, invented the term *chartjunk* to refer to any elements in a visual that are either unnecessary or irrelevant to the reader's understanding of the data or that impede or distract from a reader's ability to understand the data.

Technology enables business communicators to incorporate many features (e.g.,

colors, lines, images) into their visuals—some that will enhance a visual's message and some that will not. Charley Kyd from the ExcelUsers blog explains that the distinction between visual elements that enhance a visual versus those that are chartjunk is the extent to which visual elements make the reader work to understand the information.



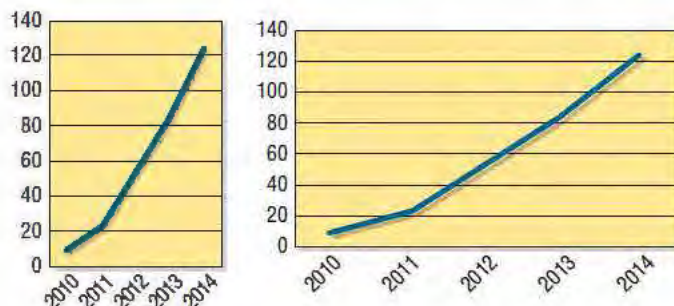
Source: Reprinted with permission of Charley Kyd, ExcelUser.com.

To illustrate, Kyd presents variations on the same visual about Juicy sales from the April 24, 2012, edition of *The Wall Street Journal*. The one on the left contains chartjunk; the one on the right is Kyd's rendering of the visual minus the chartjunk.

Kyd acknowledges that neither visual makes the data impossible to understand; what makes the elements chartjunk in the visual on the left is that the visual elements "act as noise where your readers need silence." That is, the shape of the bite in the apple appears to alter the shape of the bars in the chart, making the reader look twice to see that the bars, indeed, retain their shape. Is this a lot of work for the reader? No, but why make your reader work at all? When you create visuals, just as when you create text, all elements should enhance your reader's understanding of your message.

Sources: Charley Kyd, "Oh, No! Chart Junk from *The Wall Street Journal*," *ExcelUser Blog: Insight for Business Users of Microsoft Excel*, WordPress 12 May 2012, Web, 12 July 2012; Jessica E. Vascellaro and Ian Sherr, "Apple Rides iPhone Frenzy: Quarterly Profit Nearly Doubles as Tech Giant Taps China, New Markets," *The Wall Street Journal*, Dow Jones & Company, Inc., 24 Apr. 2012, Web, 24 May 2013.

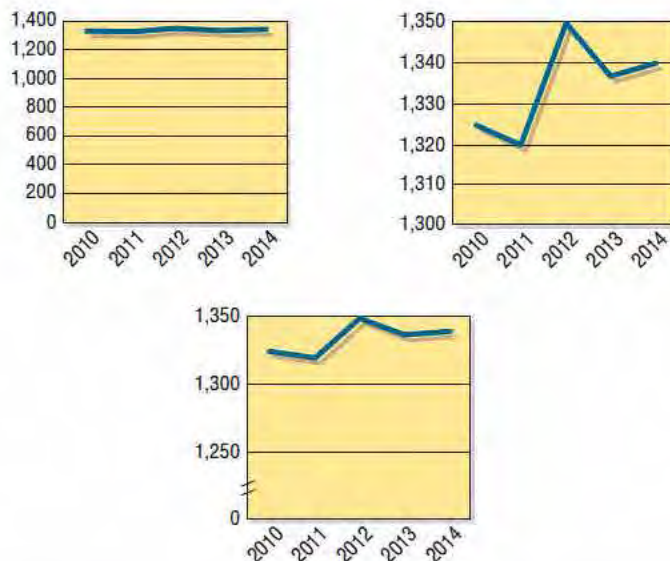
Scale distortion occurs when a visual is stretched excessively horizontally or vertically to change the meaning it conveys to the reader. Expanding a scale can change the appearance of the line. For example, if the values on a chart are



plotted one-half unit apart, changes appear to be much more dramatic. Determining the distances that present the most accurate picture is a matter of judgment. Notice the different looks of the visuals at the bottom of the left column when they are stretched vertically and horizontally.

Finally, another type of scale error is the **missing zero beginning** of the series. For accuracy you should begin the scale at zero. But when all the information shown in the chart has high values, it is awkward to show the entire scale from zero to the highest value. For example, if the quantities compared range from 1,320 to 1,350 and the chart shows the entire area from zero to 1,350, the line showing these quantities would be almost straight and very high on the chart. Your solution in this case is not to begin the scale at a high number (say 1,300),

which would distort the information, but to begin at zero and show a scale break. Realize, however, that while this makes the differences easier to see, it does exaggerate the differences. You can see this effect here.



Occasionally, though, a writer needs to use his or her judgment when the guides for starting at the zero point are not practical. For example imagine a graph compares race times of three runners, all of whom have times that cluster in the 14–17 second



Computer applications enable writers to easily create a wide variety of visuals from small to huge data sets.

range. Beginning the time scale at 0 would hinder the reader's interpretation of the information. In this context, presenting the data at a scale that begins at 14 efficiently present the data while maintaining integrity.

Errors of format come in a wide variety. Some of the more common ones are choice of wrong chart type, distracting use of grids and shading, misuse of typeface, and



communication matters

The Periodic Table of Visualization Methods

This chapter presents many of the most common options for using visuals to present your text and data. The Periodic Table of Visualization Methods provides an array of many other possibilities to help you choose the best visual as well. To see your options,

just visit the Web site www.visual-literacy.org/periodic_table/periodic_table.html, move your mouse over any of the visual types, and view the example that appears. As you view the many creative options, you may be tempted to choose a visual format based

on its novelty rather than its functionality. Remember, though, that the main purpose of any visual is to communicate. Choose wisely. You want your visuals to look good, but more importantly you want them to be appropriate for your message.

A PERIODIC TABLE OF VISUALIZATION METHODS

 C continuum		 Data Visualization Visual representations of quantitative data in schematic form (either with or without axes)		 Strategy Visualization The systematic use of complementary visual representations in the analysis, development, formulation, communication, and implementation of strategies in organizations.		 G graphic notation	
 Tb table		 Ca cartesian coordinates		 Metaphor Visualization Visual Metaphors position information graphically to organize and structure information. They also convey an insight about the represented information through the key characteristics of the metaphor that is employed		 Me meeting trace	
 Pi pie chart		 L line chart		 Compound Visualization The complementary use of different graphic representation forms in one single schema or frame		 Mm metro map	
 B bar chart		 Ac area chart		 R radar chart cobweb		 Tm temple	
 Hi histogram		 Sc scatterplot		 Pa parallel coordinates		 St story template	
 Tk tally box plot		 Sp spectrogram		 Hy hyperbolic tree		 Tr tree	
 Cy cycle diagram		 T time-line		 Ve venn diagram		 Et eternity	
 Fl flow chart		 Mi mindmap		 Sq square of opposition		 Cc concentric circles	
 Cl clustering		 Lc layer chart		 Py pyramid technique		 Ar argument slide	
 Ce cause-effect chain		 Ti toulmin map		 Sw swim lane diagram		 Gc gantt chart	
 Fo force field diagram		 Ib ibc argumentation map		 Pm perspectives diagram		 D diamond diagram	
 Pr process event chart		 Kn knowledge map		 Pr parameter ruler		 Lm life map	
 TECHNOLOGY ROADMAP							
 Cy Process Visualization		Note: Depending on your location and connection speed it can take some time to © Ralph Lengler & Martin J. Eppler, www.visual-literacy.org					
 Hy Structure Visualization		 Su supply demand curve		 Pc performance charting		 St strategy map	
 Oc organisation chart		 Ho house of quality		 Fd feedback diagram		 Ft failure tree	
 Mq magic quadrant		 Ld life-cycle diagram		 Po porter's five forces		 S s-cycle	
 Sm stakeholder map		 Is ishikawa diagram		 Tc technology roadmap		 Ed edgeworth box	
 Pf portfolio diagram		 Sg strategic game board		 Mz mintzberg's organigram		 Z zwicky's morphological box	
 Ad affinity diagram		 De decision discovery diagram		 Bm bcg matrix		 Stc strategy canvas	
 Vc value chain		 Hy hypo-cycle		 Sr stakeholder rating map		 Ta taps	
 Sd spray diagram		 Su supply demand curve		 Pc performance charting		 St strategy map	
 Ho house of quality		 Fd feedback diagram		 Ft failure tree		 Mq magic quadrant	
 Ld life-cycle diagram		 Po porter's five forces		 S s-cycle		 Sm stakeholder map	
 Is ishikawa diagram		 Tc technology roadmap		 Ed edgeworth box		 Pf portfolio diagram	
 Sg strategic game board		 Mz mintzberg's organigram		 Z zwicky's morphological box		 Ad affinity diagram	
 De decision discovery diagram		 Bm bcg matrix		 Stc strategy canvas		 Vc value chain	
 Hy hypo-cycle		 Sr stakeholder rating map		 Ta taps		 Sd spray diagram	

Source: Ralph Lengler and Martin J. Eppler, www.visual-literacy.org. Reprinted with permission of Professor Dr. Martin Eppler.

communication matters

Practicing Visual Ethics

As you have learned in this chapter, visuals can serve several useful purposes for the business writer. However, the writer needs to be accountable in using visuals to present images that in the eye and mind of the reader communicate accurately and completely. To do this, the careful writer pays attention to both the design and content of the visual. These are particularly important, for readers often skim text but read the visuals. Research shows that people remember images much better and longer than text.

Donna Kienzler offers the following guides to help you evaluate the visuals you use:

- Does the visual's design create accurate expectations?
- Does the story told match the data?
- Is the implied message congruent with the actual message?
- Will the impact of the visual on your audience be appropriate?
- Does the visual convey all critical information free of distortion?
- Are the data depicted accurately?

We would also add that ethical use of visuals requires you to cite the source of any data that you did not produce. In addition, you must get permission to use visuals that you do not create or own—including those found through Web searches.

Source: Adapted from Donna S. Kienzler, "Visual Ethics," *Journal of Business Communication* 34 (1997): 171–87, print.

problems with labels. For example, if a company used pie charts to compare expenses from one year to the next, readers might be tempted to draw conclusions that would be inappropriate because, although the pies would both represent 100 percent of the expenses, the size of the business and the expenses may have grown or shrunk drastically in a year's time. If one piece of the pie is colored or shaded in such a way as to make it stand out from the others, it could mislead readers. And, of course, small type or unlabeled, inconsistently labeled, or inappropriately labeled visuals confuse readers.

Another ethical challenge is accounting for the context. For example, as we have discussed, the number and size of visuals should be proportionate to the importance of the topic and appropriate for the emphasis a topic deserves.

avoiding other ethical problems There are other ethical issues to consider. Writers need to be careful when choosing the information to represent and the visual elements to represent it. One area writers need to watch is appropriate selection of the contents. Are people or things over- or underrepresented? Are the numbers of men and women appropriate for the context? Are their ages appropriate? Is ethnicity represented appropriately? Have colors been used appropriately and not to evoke or manipulate emotions? Writers need to carefully select and design visuals to maintain integrity.

LO 3-6 Place and interpret visuals effectively.

PLACING AND INTERPRETING THE VISUALS

For the best communication effect, you should place each visual near the place where it is discussed. Exactly where on the page you should place it, however, should be determined by its size. If the visual is small, you should place it within the text that discusses it. If it is a full page, you should place it on the page following the first reference to the information it covers.

Some writers like to place all visuals at the end of a document, usually in the appendix. This arrangement may save time in preparing the document, but it makes the readers' task more difficult because they have to flip through pages every time they want to see a visual. Therefore, place visuals where they are most helpful to the reader.

That said, sometimes you may have a visual that is necessary for completeness but is not discussed in the document (e.g., a print-out of an online survey or list of questions from an interview cited in a report). Or you may have summary charts or tables that apply to the entire document but to no specific place

in it. When such visuals are appropriate, you should place them in an appendix, and you should refer to the appendix at an appropriate point in the document.

Visuals communicate most effectively when the readers view them at the right point in their reading. Thus, you should tell the readers when to look at a visual and what to see. Of the many wordings used for this purpose, these are the most common:

As Figure 4 shows,

. . . , indicated in Figure 4,

. . . , as a glance at Figure 4 reveals,

. . . (see Figure 4)

If your visual is carrying the primary message, as in a detailed table, you can just refer the reader to the information in the

visual, as in “As Table 1 illustrates, our increased sales over the last three years. . . .” No further explanation or discussion may be necessary.

However, sometimes the visual is part of a more detailed discussion or presentation of your data. In these cases, you will start with a summary statement that reveals the big picture. If you were discussing Exhibit 3-17 (page 54), you might say, “As Exhibit 3-17 shows, areas of the country with the lowest unemployment rate are in the High, Central, and Southern Plains states.” After presenting the figure, you would call your readers’ attention to more specific points in the visual. Then you would give the exception to the general trend, if there is one.

Your readers will appreciate well-chosen, well-designed, and well-explained visuals, and you will achieve powerful communication results.